

The scenario



- ▶ Due to more or less recent events, a new law in the UK declares the possession of any pictures showing pigs together with humans or any material related to pictures showing pigs with humans strictly illegal.
- ▶ One of our lecturers is accused of having downloaded and saved the Wikipedia page about the topic “Pig” on his university computer to a local hard drive, that also includes an image of a human with a pig.
- ▶ The tool to commit this offense was probably the Firefox browser using the “Save” feature running on Windows 8.1.
- ▶ A forensic image of the disk has been taken using “CAINE”

The Alternative scenario

- ▶ The lecturer denies all accusations; however he admits that such files exist on his computer.
- ▶ However, he claims that he did not download these files.
- ▶ Instead, he assumes that somebody else might have copied these files from an USB stick to his local hard drive while he was away from his computer.

Your Task - Investigate!

- ▶ Investigate this case
- ▶ You are given two images, each one of them belongs to one of the alternative scenarios
- ▶ Analyse both of them to determine what supports the claims in one task, and makes the claims from the other task unlikely.
- ▶ Use Autopsy!

Scenario 1

MFT = Master File Table
Updated on change of: MAC
times, filename, size, ...

scenario1 - Autopsy 4.5.0

07:53 investigator

Strings File Metadata Results Indexed Text Media Other Occurrence

28 Results

Time Created Time Size Flags(Dir) Flags(Meta) Mode UserID Group

45:39 BST	2015-10-21 11:45:39 BST	56	Allocated	Allocated	drwxrwxrwx	0	0
45:40 BST	2015-10-21 11:28:20 BST	56	Allocated	Allocated	dr-xr-xr-x	48	0
45:39 BST	2015-10-21 11:45:39 BST	545	Allocated	Allocated	rw-rw-rw-r	0	0
45:40 BST	2015-10-21 11:45:40 BST	11873	Allocated	Allocated	rw-rw-rw-r	0	0
45:40 BST	2015-10-21 11:45:40 BST	870	Allocated	Allocated	rw-rw-rw-r	0	0
45:40 BST	2015-10-21 11:45:40 BST	637	Allocated	Allocated	rw-rw-rw-r	0	0
45:40 BST	2015-10-21 11:45:40 BST	675	Allocated	Allocated	rw-rw-rw-r	0	0
45:40 BST	2015-10-21 11:45:40 BST	649	Allocated	Allocated	rw-rw-rw-r	0	0
45:40 BST	2015-10-21 11:45:40 BST	6322	Allocated	Allocated	rw-rw-rw-r	0	0
45:40 BST	2015-10-21 11:45:40 BST	7045	Allocated	Allocated	rw-rw-rw-r	0	0
45:40 BST	2015-10-21 11:45:40 BST	5246	Allocated	Allocated	rw-rw-rw-r	0	0
45:40 BST	2015-10-21 11:45:40 BST	5504	Allocated	Allocated	rw-rw-rw-r	0	0
45:39 BST	2015-10-21 11:45:39 BST	15432	Allocated	Allocated	rw-rw-rw-r	0	0
45:40 BST	2015-10-21 11:45:40 BST	7315	Allocated	Allocated	rw-rw-rw-r	0	0
45:39 BST	2015-10-21 11:45:39 BST	27344	Allocated	Allocated	rw-rw-rw-r	0	0
45:39 BST	2015-10-21 11:45:39 BST	16041	Allocated	Allocated	rw-rw-rw-r	0	0
45:39 BST	2015-10-21 11:45:39 BST	1042	Allocated	Allocated	rw-rw-rw-r	0	0

STANDARD_INFORMATION Attribute Values:
Flags: Archive
Parent MFT Entry: 0
Security ID: 268 (S-1-5-21-1981759078-776323404-742695534-1001)
Created: 2015-10-21 11:45:39.997303000 (BST)
File Modified: 2015-10-21 11:45:40.000375800 (BST)
MFT Modified: 2015-10-21 11:45:39.997303000 (BST)
Accessed: 2015-10-21 11:45:39.997303000 (BST)

\$FILE_NAME Attribute Values:
Flags: Archive
Name: 220px-Kelgris1909b.jpg
Parent MFT Entry: 41 Sequence: 2
Allocated Size: 0 Actual Size: 0
Created: 2015-10-21 11:45:39.997303000 (BST)
File Modified: 2015-10-21 11:45:39.997303000 (BST)
MFT Modified: 2015-10-21 11:45:39.997303000 (BST)
Accessed: 2015-10-21 11:45:39.997303000 (BST)

Attributes:
Type: \$STANDARD_INFORMATION (16-0) Name: N/A Resident size: 72
Type: \$FILE_NAME (48-2) Name: N/A Resident size: 110

MAC times are consistent and have 100ns resolution

It is normal that access time is not updated (could be delayed/disabled)

Scenario 2

100ns resolution not present

MAC times indicate "copy" (only M is preserved)

MAC times indicated "copy" (nothing preserved)

\$STANDARD_INFORMATION Attribute Values:
Flags: Archive
Owner ID: 0
Security ID: 269 (S-1-5-21-1981759078-776323404-742695534-1001)
Created: 2015-10-21 11:48:21.801175800 (BST)
Modified: 2015-10-21 11:45:24.000000000 (BST)
MFT Modified: 2015-10-21 11:48:21.925204900 (BST)
Accessed: 2015-10-21 11:48:21.801175800 (BST)

\$FILE_NAME Attribute Values:
Flags: Archive
Name: 220px-Kelgris1909b.jpg
Parent MFT Entry: 41 Sequence: 1
Allocated Size: 16384 Actual Size: 0
Created: 2015-10-21 11:48:21.801175800 (BST)
File Modified: 2015-10-21 11:48:21.801175800 (BST)
MFT Modified: 2015-10-21 11:48:21.801175800 (BST)
Accessed: 2015-10-21 11:48:21.801175800 (BST)

Attributes:
Type: \$STANDARD_INFORMATION (16-0) Name: N/A Resident size: 72
Type: \$FILE_NAME (48-2) Name: N/A Resident size: 110

Time	Flags(Meta)	Mode	UserID	GroupID
10-21 11:48:21 BST	56	Allocated	Allocated	drwxrwxrwx
10-21 11:28:40 BST	56	Allocated	Allocated	drwxrwxrwx
10-21 11:48:21 BST	545	Allocated	Allocated	drwxrwxrwx
10-21 11:48:21 BST	11673	Allocated	Allocated	drwxrwxrwx
10-21 11:48:21 BST	870	Allocated	Allocated	drwxrwxrwx
10-21 11:48:21 BST	837	Allocated	Allocated	drwxrwxrwx
10-21 11:48:21 BST	1075	Allocated	Allocated	drwxrwxrwx
10-21 11:48:21 BST	27344	Allocated	Allocated	drwxrwxrwx
10-21 11:48:21 BST	16041	Allocated	Allocated	drwxrwxrwx
10-31 11:48:21 BST	1047	Allocated	Allocated	drwxrwxrwx